HOI: Retry-safe and Accessible Guest and Staff Pages

One tap, one job: Idempotency-Key for guest requests and staff actions, and WCAG 2.1 AA fixes with an automated accessibility gate.

Saumya Goswami · Technical documentation · October 2026

Project: github.com/Ashu-Stark/Hospitality-

Contents

1. Summary

2. Retry safety

3. Accessibility

4. Tests

5. Limitations

6. Interview questions

1 Summary

HOI is Team Innov8's hotel operations platform. Guests use a room-QR page to ask for towels, food or help, and staff use a phone app to accept and finish jobs. This contribution makes those two pages **safe to retry**: one tap creates one job, and a retried staff action never turns success into an error. It also makes them **usable by everyone**, measured against WCAG 2.1 AA, with an automated check that keeps them that way.

5 → 1
jobs from six parallel taps of one request

28 → 0
axe accessibility violations (10 page states)

24 → 0
Hindi text nodes read with the wrong voice

325 → 336
platform API tests passing

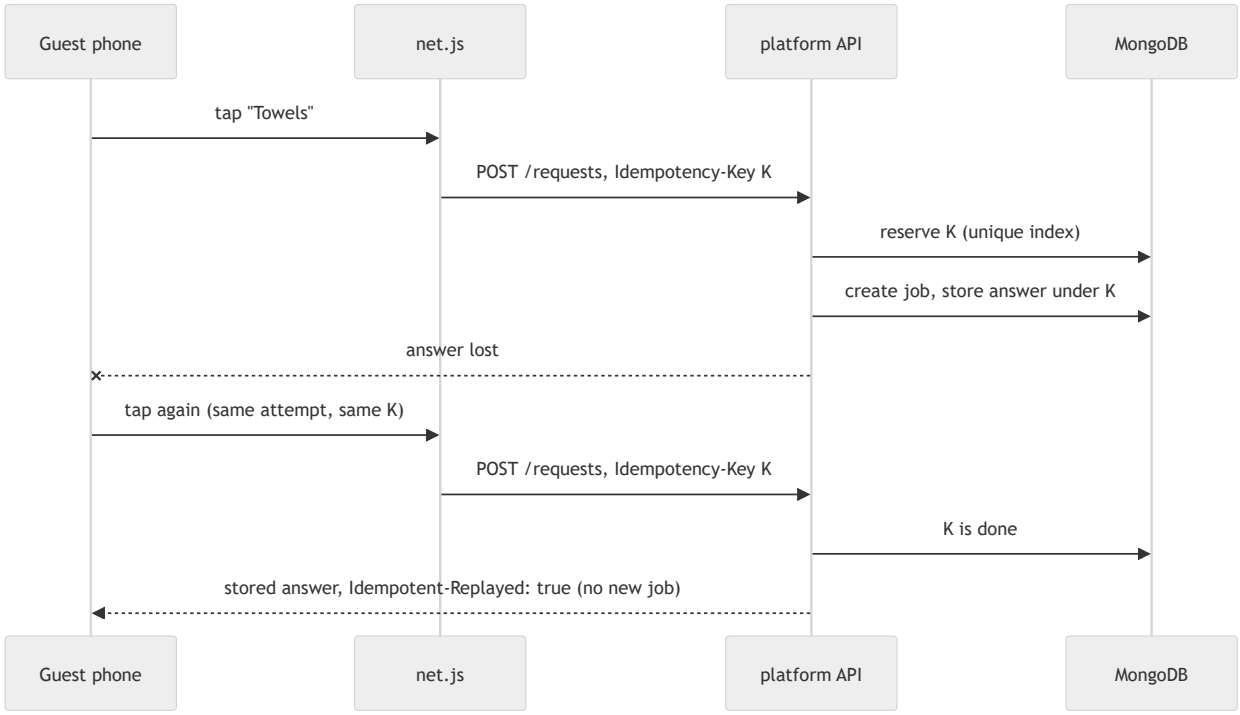
2 Retry safety

What was wrong

Situation	Before
Guest's request succeeds, the answer is lost (hotel Wi-Fi, a gateway 504 from a sleeping server), guest taps again	**Two jobs** and two staff offers
Six parallel taps of the same request	**5 jobs**, stopped only by the per-room limit of 5
Worker's "accept" succeeds, answer lost, phone retries	**409 "offer no longer available"**: the worker is told they failed
Supply request sent 3 times	**3 requisitions**
`net.js` retried any POST after 502/503/504 ("the app never saw it")	Wrong for a 504: the app may still be working

Each row was first reproduced by a failing test in `tests/test_idempotency.py`.

The fix: Idempotency-Key



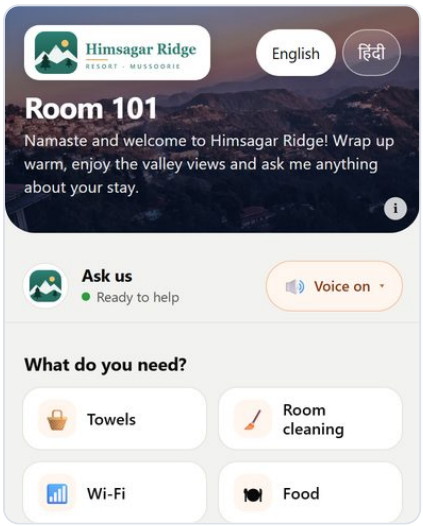
- **Same key, same body**: the stored answer again. **Still running**: 409, safe to retry. **Different body**: 422. **Action failed**: key released, so a corrected retry works.
- Keys are scoped by room token or user and endpoint, and expire after 24 h (TTL index).
- Used on guest requests, the concierge's Confirm, job accept/decline/start/complete and supply requests. **Optional header**: old clients behave exactly as before.
- `HOINet.key()` keeps one key per user attempt until the server answers, and automatic POST retries now happen only with a key.

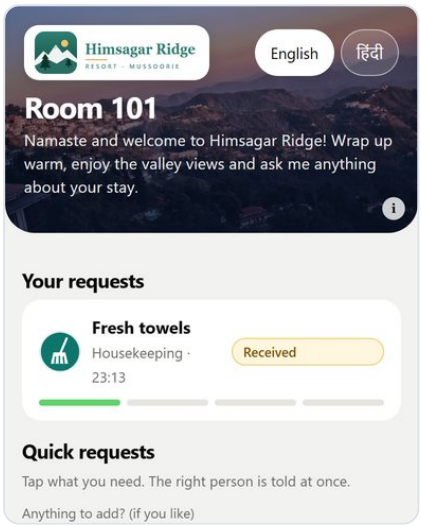
3 Accessibility

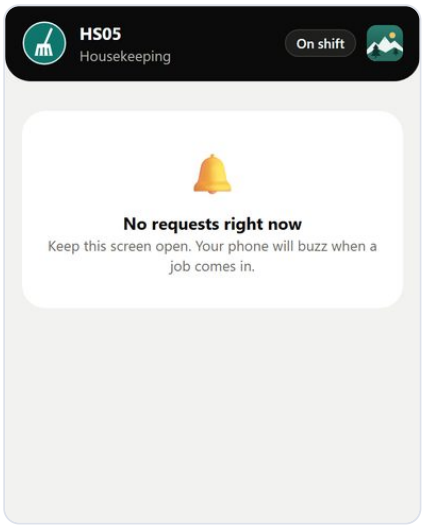
Measured on 10 page states at phone size (390 x 844). Guest: Ask, Hotel info, My requests, Check-out and Hotel info in Hindi. Staff: sign-in, Today, Jobs, Supplies and Me. Four checks run on each: axe-core 4.10 (WCAG 2.0/2.1 A + AA), controls under 44 x 44 px, Hindi text without `lang="hi"`, and whether "Request sent" reaches a `role="status"` region. "Before" is main (`a679ba4`).

Check	Before	After
axe violations	28 elements (aria-prohibited-attr, serious)	**0**
Controls under 44 x 44 px	5 controls	**0**
Hindi text without lang="hi"	24 text nodes	**0**
"Request sent" announced	No	**Yes**

Problem	Who it hurt	Fix
Veg / non-veg marks: `aria-label` on an `<i>` without a role is ignored	A blind vegetarian guest	`role="img"`
Toasts ("Request sent", "Job done") only visual	Screen-reader users never heard the confirmation	Polite `role="status"` region
Hindi labels on English pages had no `lang`	Read with an English voice	`lang="hi"` / `lang="en"`
Buttons 40 px and 22 px, checkboxes 20 px	Mis-taps on phones	44 px tap areas (the small dot stays small on screen)
Staff tab icons read aloud ("high voltage, Jobs")	Screen-reader users	`aria-hidden="true"`







Keeping it fixed: `tests/test_accessibility.py` serves the app, runs the audit and fails on any problem. Undoing the `role="img"` fix made it fail at once. CI installs Node 22 and axe-core for it.

4 Tests

Suite	Before	After
platform_api (offline, mongomock)	325 passed	**336 passed** (+10 retry-safety, +1 accessibility gate)
test_idempotency.py repeated 15 times		15 / 15 passed

5 Limitations

- The parallel test runs on mongomock, not under load on a real MongoDB Atlas cluster.
- If a guest closes the page during a lost-answer retry, the next tap is a new attempt.
- Automated checks cover only part of WCAG; a TalkBack/NVDA session with real users is still needed.
- 44 px is WCAG 2.5.5 (AAA) and phone guidance, stricter than the AA minimum, chosen because everyone here is on a phone.

6 Interview questions

Why not just disable the button while sending?

It already was. It cannot help when the answer is lost after the server did the work. Only the server can recognise the retry, and the key is how it does.

Why does a running key return 409 and not wait?

Holding the request open ties up a worker and can time out at the gateway again. 409 says "try again shortly", and the client keeps the same key, so the next try replays the answer.

Why release the key when the action fails?

A failure did nothing. Storing it would replay the error even after the guest fixed the problem.

Why role="status" and not role="alert" for "Request sent"?

It is good news, not an error. "Polite" waits for the screen reader to finish speaking; alerts interrupt.

What does lang="hi" change?

Screen readers switch to a Hindi voice for that text. Without it, Devanagari is read with English rules or skipped.